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Problem

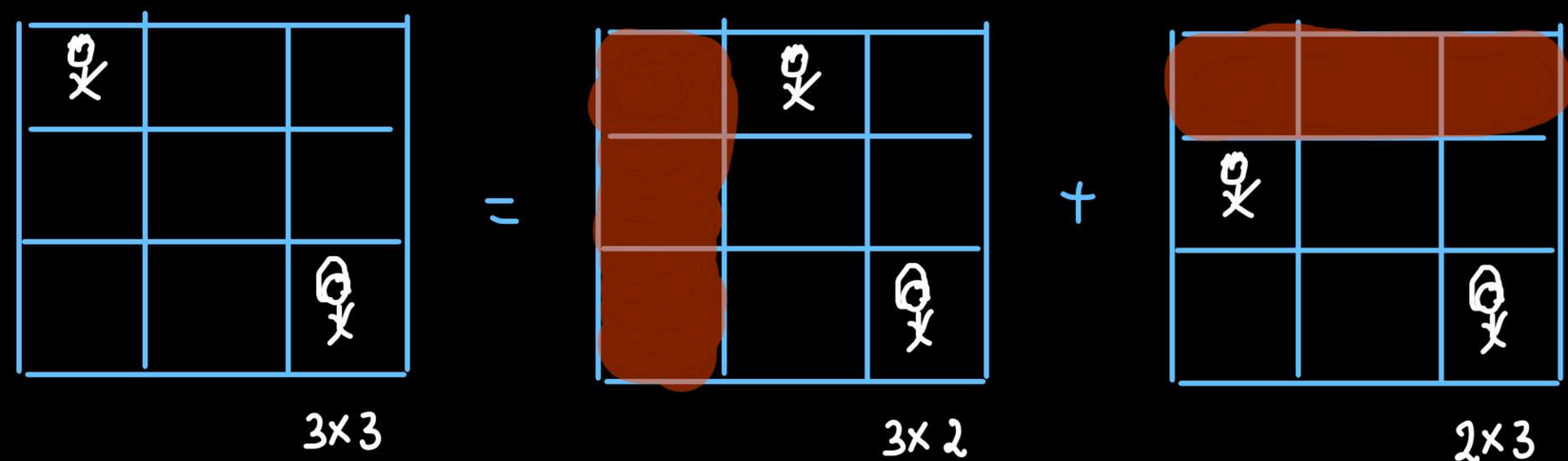
Down, Right

Number of Paths

memoization

Time = O(mn)

Space = O(mn)



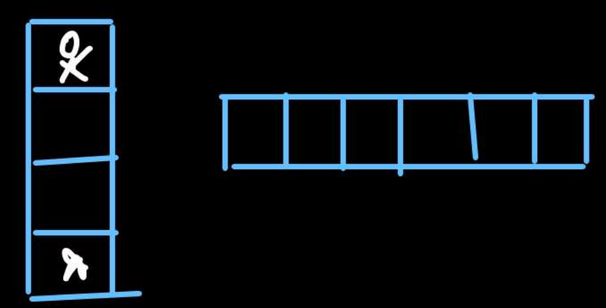
paths(m,n) = paths(m, n-1) + paths(m-1, n)

Anuj to Madhav

RRDD DRRD

RDRD DRDR

RDDR DDRR



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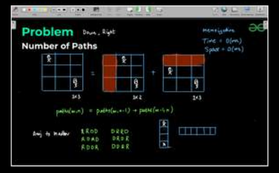
Tabulation (Asli DP)

Bottom Up DP

Iterative DP



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Problem TAYYARI5000

Nth Fibonacci Number (Tabulation)

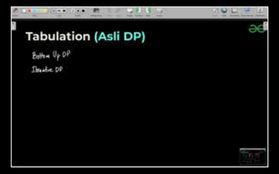
fibo(7)

	0	1	2	3	4	5	6	7
arr	0	1	1	2	3	5	8	13

$$arr[i] = arr[i-1] + arr[n-2]$$

O(n) time

O(n) space



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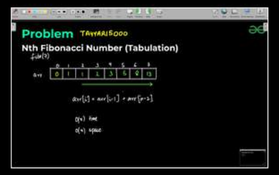
Homework:

1) n^{th} stair (tabulation)

2) House Robber 2 (Tabulation)

3) n^{th} tribonacci number (LC) (tabulation)



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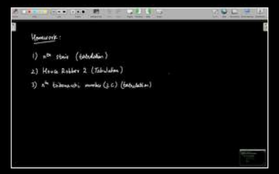
Problem

Stickler Thief (Tabulation)

money	6	10	1	2	15
dp	6	10	10	12	25

dp[i] me i^{th} ghar tak ki maximum loot store hogi

$$dp[i] = \max(money[i] + dp[i-2],\ dp[i-1])$$



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Problem

Min Cost Climbing Stairs

bishal
↓
1 or 2
jump

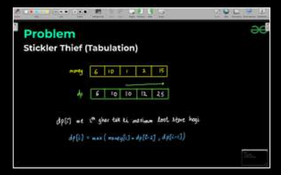
$$dp[i] = cost[i] + \min(dp[i-1], dp[i-2])$$

	0	1	2	3	4	5
cost	1	2	10	1	1	5

dp	1	2	11	3	4	8

min

dp[i] will store min cost upto that stair including that stair.



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Problem

Number of Paths (Tabulation)

$m+n-2 C_{m-1}$

$O(m)$ time

$O(1)$ space

$m=3,\ n=3$

	0	1	2
0	1	1	1
1	1	2	3
2	1	3	6

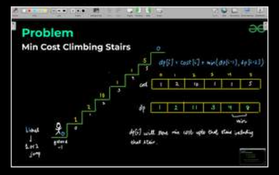
dp

$m=4,\ n=7$

1	1	1	1	1	1	1
1	2	3	4	5	6	7
1	3	6	10	15	21	28
1	4	10	20	35	56	84

Time $= O(mn)$ Space $= O(mn)$

dp[i][j] is storing no. of paths from start to (i,j)



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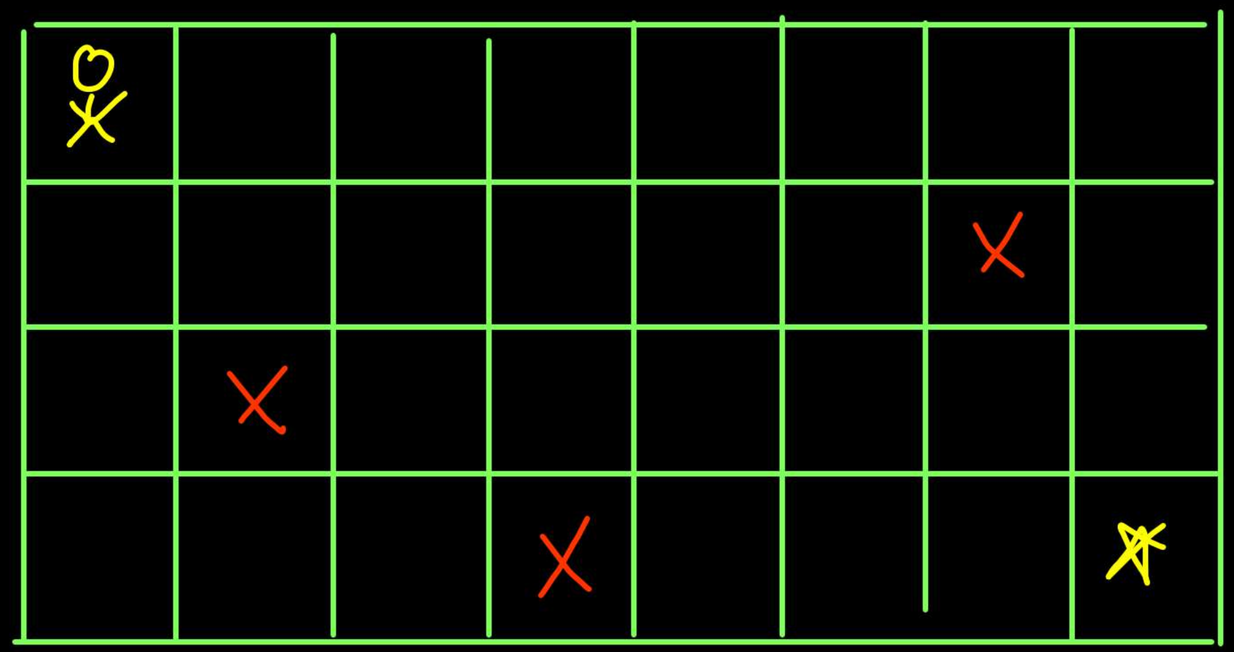
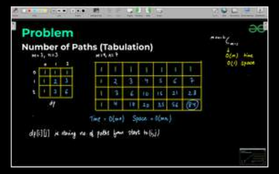
HW : Unique Paths 2





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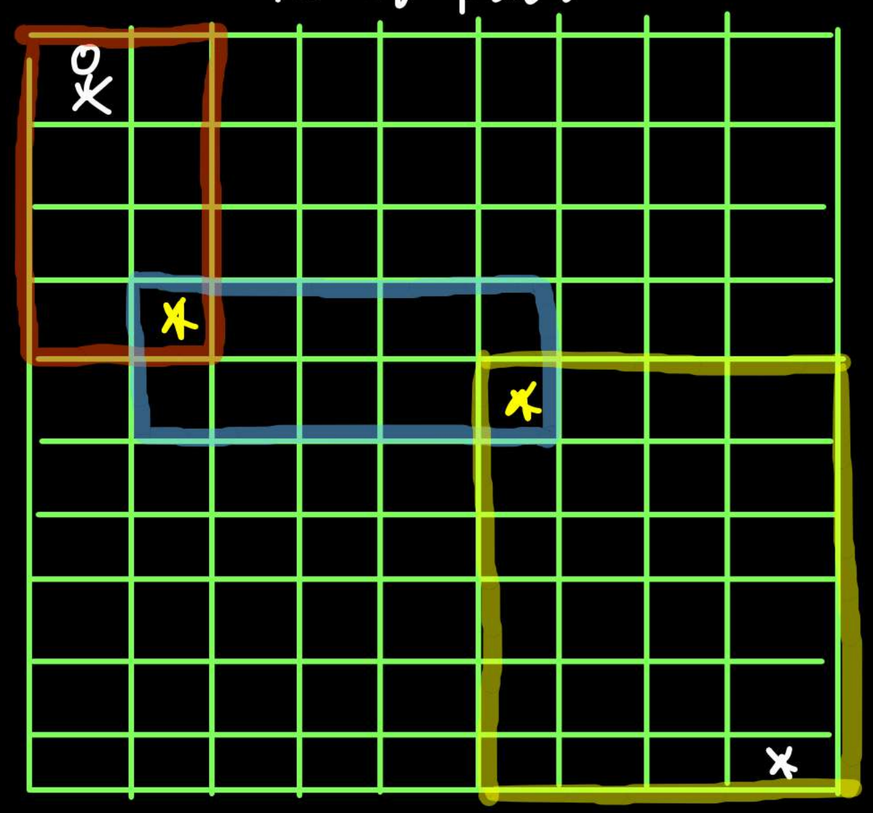
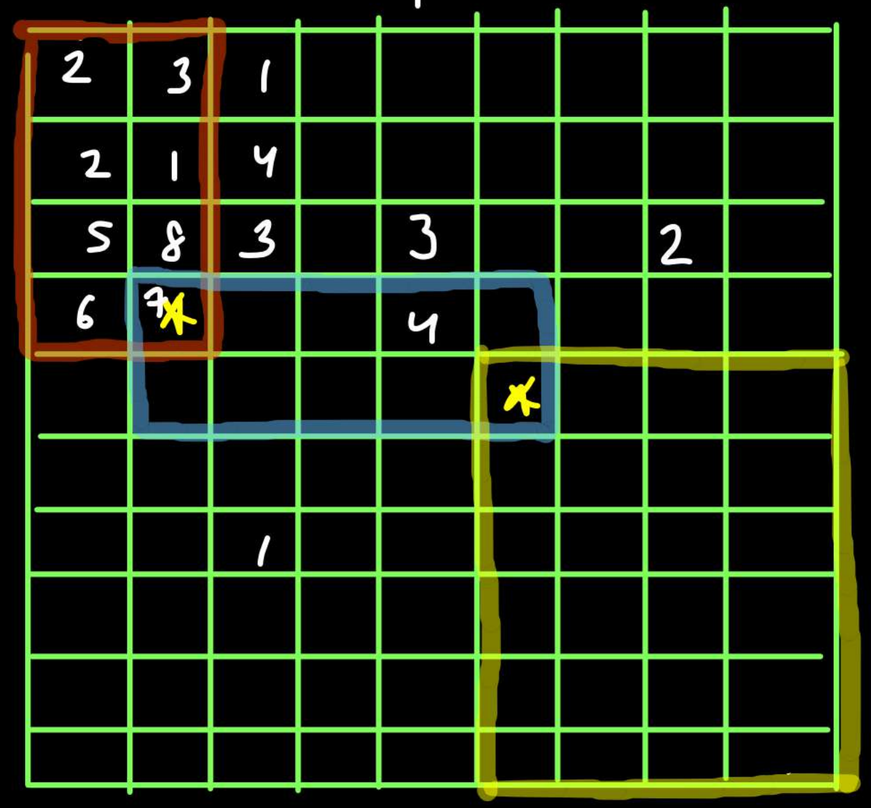
Grid Problems

1) Passing through some mandatory checkpoints

no of path

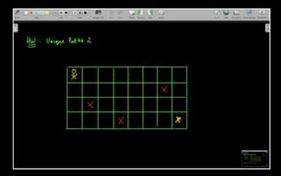


min path sum



$p1 \times p2 \times p3 = ans$

$sum1 + sum2 + sum3 .. = sum$



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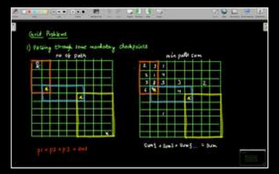
Problem

Minimum ~~Cost~~ Sum Path (Tabulation)

1	3	1
1	5	1
4	2	1

grid

1	4	5
2	7	6
6	8	7

dp